

PROCESS SPECIFICATION

PS-2025-004 Rev A

Surface Finish Requirements for Machined Components

1. SCOPE

This specification defines surface roughness requirements and measurement methods for all machined components produced under Meridian Precision Manufacturing quality system. Applies to metallic and non-metallic substrates.

2. SURFACE ROUGHNESS CLASSIFICATIONS

Class	Ra (microinches)	Ra (micrometers)	Typical Application
A - Mirror	8 max	0.2 max	Optical, sealing surfaces
B - Fine	16 max	0.4 max	Hydraulic components, bearing surfaces
C - Precision	32 max	0.8 max	Mating surfaces, press fits
D - Standard	63 max	1.6 max	General machined surfaces
E - Rough	125 max	3.2 max	Non-critical, as-machined
F - As-Cast	250 max	6.3 max	Unmachined casting surfaces

3. MEASUREMENT

Surface roughness shall be measured using a contact profilometer with a 0.030 inch cutoff length. Measurements shall be taken perpendicular to the lay direction. Minimum three measurements per surface, reported as the average.

4. DOCUMENTATION

Surface finish measurements shall be recorded on the First Article Inspection report and included in the quality data package for each lot.

Effective Date: January 10, 2025 | Approved by: K. Yamamoto, Process Eng.